Uses of Index Numbers

AQA · Macroeconomics · 2.1.3 *Updated 21 August 2026*

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> **Specification Coverage:** AQA unit 2.1.3 - Uses of Index Numbers. Students should be able to explain how index numbers are calculated and interpreted, including the base year and the use of weights. Students should also understand how index numbers are used to measure changes in the price level and other economic variables.

What Are Index Numbers?

> **Index number:** A number used to express the value of a variable relative to a chosen **base value**, which is set equal to **100**. Index numbers make it easy to compare how a variable has changed over time and to express those changes in percentage terms.

Economists use index numbers because they turn awkward raw figures - such as thousands of individual prices, or GDP measured in billions of pounds - into a single, easy-to-read series. Instead of comparing £200,000 with £230,000, we can simply compare an index of 100 with an index of 115.

Index numbers are not only used for prices. They are also used to track changes in **real GDP**, **wages**, **productivity** and share prices (such as the FTSE 100), among many other economic variables.

The Base Year

Base year: The reference year against which all other years are compared. The value of the variable in the base year is set to **100**, and every other year is expressed relative to it.

Calculating an Index Number

$$\text{Index Number} = \frac{\text{Value in Current Year}}{\text{Value in Base Year}} \times 100$$

Worked Example: Building a House Price Index

Suppose the average house price in a town is recorded over three years, with **2022 chosen as the base year**:

Year	Average House Price	Index Number (2022 = 100)
2022 (base)	£200,000	$\frac{200{,}000}{200{,}000} \times 100 = 100$
2023	£210,000	$\frac{210{,}000}{200{,}000} \times 100 = 105$
2024	£230,000	$\frac{230{,}000}{200{,}000} \times 100 = 115$

An index of **115** in 2024 tells us that house prices are **15% higher** than in the base year of 2022.

Interpreting Index Numbers

- An index of **100** represents the base year value.
- An index **above 100** means the variable has risen since the base year (e.g. 115 = 15% higher).
- An index **below 100** means the variable has fallen since the base year (e.g. 92 = 8% lower).

> EXAM TIP
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> **Only read a percentage change straight off the base.** You can read a percentage change directly from an index number *only* when comparing with the base year (100). To find the percentage change between two *non-base* years, you must use the standard percentage change formula.

$$\backslash\% \text{ Change} = \frac{\text{New Index} - \text{Old Index}}{\text{Old Index}} \times 100$$

For example, the percentage change in house prices between 2023 and 2024 is **not** simply $115 - 105 = 10\%$. It is:

$$\frac{115 - 105}{105} \times 100 = 9.5\%$$

How CPI and Inflation Are Calculated

> **Consumer Prices Index (CPI):** The main measure of the price level in the UK. It is a **weighted price index** that tracks the average change in the prices of a representative "basket" of goods and services bought by households. The annual percentage change in the CPI is the headline **rate of inflation**.

The Full Process

The Office for National Statistics (ONS) calculates the CPI through the following steps:

1. **Survey household spending.** The *Living Costs and Food Survey* asks thousands of households to record what they buy, revealing the spending patterns of a typical household.
2. **Create the basket of goods and services.** A representative basket of around 700 items is chosen. It is **updated each year** so that it reflects changing tastes and technology, with items added and removed over time.
3. **Assign weights.** Each item is given a **weight** based on the proportion of household spending it accounts for. Items that households spend more on (such as housing and transport) are given a larger weight.
4. **Collect prices.** The ONS collects around 180,000 prices for these items every month, from retailers across the country and online.
5. **Convert prices to a price index.** Each item's price is expressed as an index number relative to the base year (base = 100).
6. **Calculate the weighted index.** Each price index is multiplied by its weight; the results are added together and divided by the total weight to give the CPI.

7. **Calculate inflation.** The percentage change in the CPI over 12 months gives the rate of inflation.

Why Weighting Matters

Weighting ensures that the CPI reflects the **importance** of each item in a typical household's budget. A large price rise in a heavily-weighted item (such as petrol or rent) has a much bigger effect on the CPI than the same rise in a lightly-weighted item (such as tea bags), because households spend far more on the former.

$$\text{Weighted Price Index (CPI)} = \frac{\sum(\text{Weight} \times \text{Price Index})}{\sum \text{Weights}}$$

Worked Example: Calculating a Weighted CPI

Consider a simplified basket with three spending categories. Weights are given out of 1,000 (a common convention), and each category's price index is measured relative to the base year:

Category	Weight	Price Index (base = 100)	Weight × Index
Housing	500	106	53,000
Transport	300	110	33,000
Food	200	104	20,800
Total	1,000	—	106,800

Dividing the total weighted value by the total weight gives the CPI:

$$\text{CPI} = \frac{106{,}800}{1{,}000} = 106.8$$

A CPI of **106.8** means that, on average, prices across the basket are **6.8% higher** than in the base year. Notice how the heavily-weighted housing category pulls the overall index towards its own value.

From CPI to the Inflation Rate

Inflation: A sustained rise in the general price level. It is measured as the **annual percentage change** in the CPI.

$$\text{Inflation Rate} = \frac{\text{CPI this year} - \text{CPI last year}}{\text{CPI last year}} \times 100$$

Worked Example: Calculating the Inflation Rate

Suppose the CPI was **103.2** last year and **106.8** this year:

CPI last year	103.2
CPI this year	106.8
Change in CPI	106.8 − 103.2 = 3.6
Inflation Rate	$\frac{3.6}{103.2} \times 100 = 3.5\%$

EXAM TIP

Falling inflation is not falling prices. If the inflation rate falls from 3.5% to 2%, prices are still *rising* – just more slowly. This is **disinflation**. Prices only fall when the inflation rate turns negative, which is **deflation**.